Note

- The events E and E' are mutually exclusive. (how?)
- The probability of E + the probability of E' =1. Also E and E' are mutually exclusive and exhaustive.
- Since $P(E)+P(E')=1$, if you know any one of them, you can find the other.

Progress Check

Which among the following are mutually exclusive?

Sl.No.	Trial	Event 1	Event 2
1	Roll a dice	getting a 5	getting an odd number
2	Roll a dice	getting a 5	getting an even number
3	Draw a card from a standard pack	getting a Spade Card	getting a black
4	Draw a card from a standard pack	getting a Picture Card	getting a 5
5	Draw a card from a standard pack	getting a Heart Card	getting a 7

Example 9.4

The probability that it will rain tomorrow is $\frac{91}{100}$. What is the probability that it will not rain tomorrow?

Solution

Let E be the event that it will rain tomorrow. Then E' is the event that it will not rain tomorrow.

Since $P(E) = 0.91$, we have $P(E') = 1-0.91$ (how?)

$= 0.09$

Therefore, the probability that it will not rain tomorrow

$= 0.09$

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